

Backup Infrastructure — Target Design v4 (Definitive)

Corrected for hardware reality — NTFS-local + immutable object storage · no ReFS · supersedes v1–v3

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What changed from v3 & why. v3 proposed reformatting the SAN/Synology to **ReFS** for block-clone efficiency. On verification that is the **wrong call for this hardware**: ReFS block-clone on a 25 TB+ repository is memory-hungry, and **BK01 has only 16 GB RAM** — the #1 cause of the ReFS metadata corruption that gives ReFS-on-iSCSI a bad reputation. T-Net's instinct to stay NTFS is defensible here. **v4 keeps local storage NTFS and moves archive/scaling/immutability to object storage** — which needs no ReFS and no RAM headroom, and is a stronger design regardless.

1. The architecture (3-2-1-1-0, no ReFS)

- Production** (vSphere / UC3200) → **Veeam BK01 (12.3.2)** →
- ① **Primary** → **Synology (F:, NTFS)** all VMs · fwd-incr + weekly synthetic full · **short retention** (~30 days + 1–2 monthly) · mount server APP01 · Health Check + Threat Hunter on
 - ② **Off-site** → **Cloud Connect (T-Net)** Backup Copy of all VMs (WAN accel when enabled)
 - ③ **Archive + Immutable** → **Object storage (Wasabi/B2, object-lock)** deep GFS (monthly/yearly, years) · block-deduplicated · ransomware-proof
 - ④ **Verification** → **SureBackup virtual lab** auto-boots & tests backups on schedule

- Why this shape:**
- **Local stays NTFS, short retention.** NTFS holds a handful of full-size points fine; you only hit the wall keeping a year locally — which you won't. No risky reformat, no RAM constraint.
 - **Object storage carries the heavy lifting** that ReFS was meant to: long GFS retention (block-deduplicated = cheap, scales infinitely — no 5.8 TB-per-full problem) **and** immutability (object-lock = ransomware-proof) **and** a second off-site copy — one tier, no hardware dependency on BK01's RAM.
 - **3-2-1-1-0:** 3 copies (Synology, Cloud Connect, object storage) · 2+ media · 1 off-site · 1 immutable · 0 unverified (SureBackup).

2. The five capabilities of "immaculate"

Capability	Feature	Status
Deduplicated file-level restore	APP01 (Server 2025) mount server	Done
Off-site copy	Backup Copy → Cloud Connect (T-Net)	Seeding
Failure alerting	Email notifications	Done
Automated recovery verification	SureBackup (isolated virtual-lab boot/test) + Health Checks	Next
Ransomware — malware scan	Veeam Threat Hunter (installed w/ 12.3)	Next
Ransomware — anomaly detection	Suspicious-activity alerts (12.3)	Next
Ransomware — immutable archive + deep retention	Object storage w/ object-lock (Wasabi/B2)	Phase 2

Point-in-time / clean-room history	GFS retention → object storage tier	Phase 2
Fast recovery (RTO)	Instant VM Recovery (validate; APP01 cache)	Next
WAN throughput to off-site	WAN Acceleration (T-Net cloud-side)	T-Net

3. Build plan — ordered

Phase A — Off-site copy of FS01 RUNNING

Backup Copy → Cloud Connect, seeding over the wire (~2 weeks). WAN acceleration requested from T-Net. No action needed but monitoring.

Phase B — Verify + harden (NOW, while the seed runs) KOR

1. **Health Checks** — enable on every backup job (Storage → Advanced → Maintenance → periodic health check). Auto-detects backup corruption.
2. **Threat Hunter / inline malware scan** — enable in the Security & Compliance area (12.3). Scans backups as written.
3. **Anomaly / suspicious-activity alerts** — enable; route to KOR email.
4. **SureBackup** — build a Virtual Lab on a host with headroom → Application Group (the VMs) → SureBackup job scheduled after nightly backups (boot + ping tests) → results email KOR. Automated recovery proof.
5. **Validate Instant VM Recovery** — test-boot a small VM from backup via APP01; confirm it runs; discard.

Phase C — Object storage: immutable archive + deep GFS KOR

1. Choose provider (Wasabi or Backblaze B2) and create an account + bucket with **Object Lock enabled**.
2. Veeam → Backup Infrastructure → Backup Repositories → Add → **Object Storage** → point at the bucket → enable "**Make recent backups immutable for N days.**"
3. Wrap the Synology + object storage into a **Scale-Out Backup Repository**: Synology = performance tier, object storage = **capacity tier** (offload/archive) with immutability.
4. Enable **GFS** on the FS01 job (monthly/yearly) — it tiers to object storage, block-deduplicated, so a year of points costs little. This is the durable "go back in time for files" + ransomware-proof layer.

Phase D — Reclaim + optimise KOR

1. After the off-site seed verifies: retire the old E: reverse-incr chain (~23 TB) and old cloud FS01 copy.
2. Repurpose E: as a second local copy / staging (NTFS — no reformat).
3. Convert small-VM jobs to forward incremental; finish tidy-ups (APP003/004; RDS01 duplicate — done).

4. T-Net — informed / dependencies

Informed: 12.3.2 upgrade; APP01 mount server; ~20 TB retired VMs; the NTFS-stays / object-storage-archive design; FS01 added to off-site.

Dependencies: Veeam **licence** (confirm coverage); **Cloud Connect** WAN accelerator + tenant bandwidth cap + quota; **monitoring/alerting SLA**; **vCenter cert ownership** (2028-07-14). Note: object storage is KOR-provisioned and independent of T-Net.

KOR Structural Operations, 2026-07-23. Supersedes v1–v3. ReFS dropped after verifying BK01's 16 GB RAM makes ReFS-on-large-iSCSI unsafe; object storage delivers the same efficiency + immutability without that risk. Seed estimates based on 182 Mbps upload.